

# Solution: Supplementary Problem 4

## Chapter 10: Rotational Dynamics – Varied Motion

### 1. Preliminary Data & Conversions

All quantities must be converted to the International System (SI) before calculation:

- **Radius ( $R$ ):**  $20\text{ cm} = 0.2\text{ m}$
- **Mass ( $M$ ):**  $250\text{ g} = 0.25\text{ kg}$
- **Angular Acceleration ( $\alpha$ ):**  $60\text{ rad/s}^2$  (Constant  $\Rightarrow$  U.V.R.M.)
- **Initial State:** Starts from rest ( $t_0 = 0, \omega_0 = 0$ )
- **Moment of Inertia ( $I$ ):** Since the mass is "uniformly distributed on its circumference," the wheel is assimilated to a **thin hoop**:

$$I = MR^2 = 0.25 \times (0.2)^2 = 0.25 \times 0.04 = \mathbf{0.01\text{ kg} \cdot \text{m}^2}$$

### Question 1: Resultant Moment of External Forces

According to **Newton's Second Law for Rotation** (Fundamental Law of Dynamics):

$$\sum M_{\Delta}(\vec{F}_{ext}) = I\alpha$$

Substituting the calculated value of  $I$  and the given  $\alpha$ :

$$\sum M_{\Delta} = 0.01 \times 60 = \mathbf{0.6\text{ N} \cdot \text{m}}$$

**Resultant Moment:**  $\sum M = \mathbf{0.6\text{ N} \cdot \text{m}}$

### Question 2: Linear Acceleration at $t = 0.15\text{ s}$

A point on the circumference undergoes both tangential ( $a_t$ ) and normal ( $a_n$ ) acceleration. The total linear acceleration is the magnitude of their vector sum:  $a = \sqrt{a_t^2 + a_n^2}$ .

- **Tangential Acceleration ( $a_t$ ):** Constant throughout the motion.

$$a_t = R\alpha = 0.2 \times 60 = \mathbf{12\text{ m/s}^2}$$

- **Angular Speed at  $t = 0.15\text{ s}$ :**

$$\omega = \alpha t = 60 \times 0.15 = \mathbf{9\text{ rad/s}}$$

- **Normal (Centripetal) Acceleration ( $a_n$ ):**

$$a_n = R\omega^2 = 0.2 \times (9)^2 = 0.2 \times 81 = \mathbf{16.2\text{ m/s}^2}$$

The magnitude of the resultant linear acceleration is:

$$a = \sqrt{12^2 + 16.2^2} = \sqrt{144 + 262.44} = \sqrt{406.44} \approx \mathbf{20.16\text{ m/s}^2}$$

**Linear Acceleration:**  $a \approx \mathbf{20.16\text{ m/s}^2}$

### Question 3: Number of Turns after 2 s

First, calculate the angular displacement ( $\Delta\theta$ ) using the law of motion for U.V.R.M.:

$$\Delta\theta = \frac{1}{2}\alpha t^2 + \omega_0 t$$

Since  $\omega_0 = 0$  and  $t = 2\text{ s}$ :

$$\Delta\theta = \frac{1}{2}(60)(2)^2 = 30 \times 4 = \mathbf{120\text{ rad}}$$

To find the number of turns ( $n$ ):

$$n = \frac{\Delta\theta}{2\pi} = \frac{120}{2\pi} \approx \mathbf{19.1\text{ turns}}$$

**Number of Turns:  $n \approx 19.1\text{ rev}$**